

Class 30006 – Financial Markets and Institutions
Università Commerciale Luigi Bocconi
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Review / Exercises



Ex 1: Suppose investors can only invest in the following two assets. All investors are risk averse:

<u>Stocks</u>	<u>E(R)</u>	<u>Std.Dev.</u>
ABC	\$45	20%
XYZ	\$32	10%

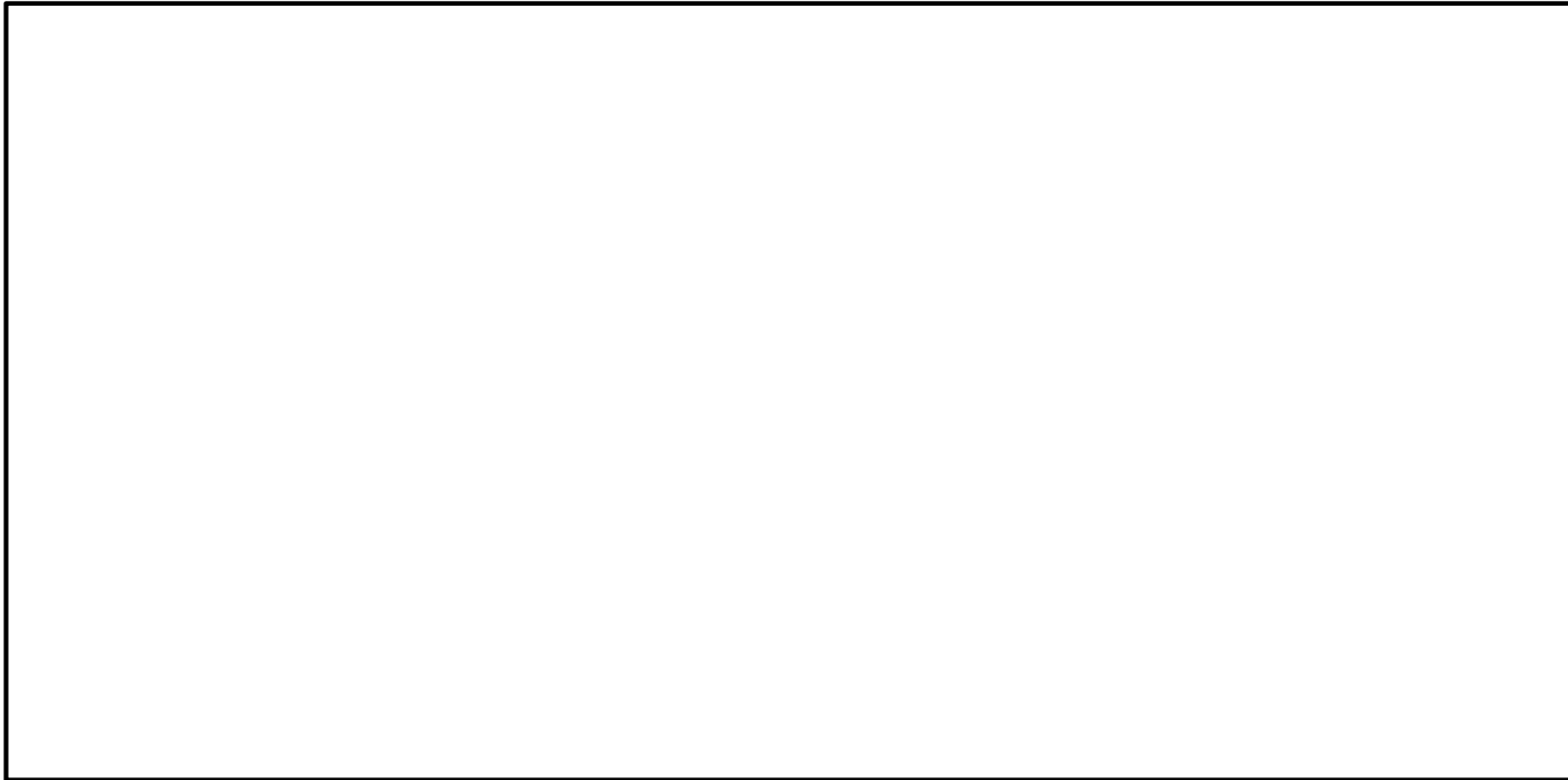
Q1: What will happen to the demand for stock ABC if its std. dev. goes up by 5 pct. points (from 20 to 25%), all else equal?

Q2: What happens if the std. dev of XYZ increases by 15 pct. points (from 10% to 25%)? Does the investor prefer stock ABC or XYZ?

Ex 2: What is the YTM on a zero-coupon bond with the face value of \$1,500, with 2 years left until maturity, and selling for \$1,200?

Q: what if it paid a 10% annual coupon?

Q: What if it paid coupon semi-annually?



Ex 3: A firm just paid an annual dividend of \$5 per share. You estimate that the dividend will grow at an annual rate of 2% from this year to the next and then at 4% forever.

Q: If the fair stock price is \$125 today, what must be the required rate of return?

Ex 4:

Q1: What is the value of a call option on the expiration date if the strike price is \$300 and the spot price of the underlying asset is \$650?

Q2: What is the strike price of a put option that expires on the same day for the same asset and has a value of \$30?

Ex 5: Two firms, XYZ and ABC, are similar in every respect. They issued bonds with the same face value and the same coupon rate.

Shortly after, XYZ's production falls unexpectedly. Firm ABC's production is unaffected.

The market expects XYZ to make all required bond payments with an 80% probability, with 20% probability debt holders receive nothing.

Q: is the price of XYZ lower or higher than ABC's? If the price of ABC's bond is \$900, what is that XYZ's bond?

Ex 6: A stock is selling today for \$55. A put option on this stock that expires today is selling for \$6.

Q: What must be the exercise price of this option?

Ex. 7: Early in 2024, a trader sold one future contract for September 30, 2024 delivery.

The contract was written for the delivery of one share. When the trader entered the market, the spot price was \$30 and the futures price for September 30 delivery was \$35.

Q1: What was the trader's gain/loss on this trade if he closed his position on September 23, 2024, when the spot price was \$54 and the futures price for September 30 delivery was \$66?

Q2: What was the trader's gain/loss on this trade if wait until September 30, when spot price is \$29?

Ex 8:

You have a stock indexed portfolio worth \$20 M today. The index's value today is \$1,000.

You enter a short position on 75 futures contracts, each with a price of \$1,200, for delivery in June 2026. The multiplier for this index futures contract is 100.

You close the position before delivery, when the index is \$900 and the futures price for June 2026 delivery is \$800.

Q: What is the net gain/loss on your entire portfolio?

Ex 9.

Firms ABC and XYZ are in the same industry.

However, ABC has a much higher PE ratio than XYZ.

Q: Which of the following can explain this?

- I. ABC's earnings are expected to rise more quickly in the future than XYZ's earnings
- II. ABC is perceived by the market to be less risky than XYZ.
- III. Both XYZ and ABC are in a fast-growing industry.
- IV. ABC's current earnings per share are higher than XYZ's current earnings per share

- a) I and II only
- b) I, II and IV only
- c) II and IV only
- d) II, III and IV only

Ex 10.

A bond with a face value of \$ 1,000 and a coupon of 5% is selling for \$1,100. Which of the following is true?

Q: Which of the following is true?

- a) the yield to maturity is 5.5%
- b) the yield to maturity is 6%
- c) none of the others
- d) the yield to maturity is less than 5%